

The probe reads the top 5 cm. Physics models the rest of the tree, and most of the city.

- The probe reads only the top ~5 cm of soil
- Roots drink 20–30 cm down
- Physics models **unmeasured trees** from forecast + nearby sensors
- A few sensors cover many trees
- **Young trees: dense sensors** (high risk, citizen-verifiable)
- **Old trees: sparse** (deep roots, they survive)
- **Honest:** physics estimates, calibrates as sensors deploy